WHERE AUTONOMY MEETS PRACTICAL DELIVERY

As autonomy continues to shape the future of intelligent systems, the IEEE SMC AICA Challenge 2026 invites teams to develop innovative solutions for collaborative autonomous systems. The 2026 competition will be featured in Bellevue, WA, USA, from October 4 to October 7. Using Quanser Interactive Labs (QLabs), teams will simulate a realistic logistics scenario in which the QCar2 ground vehicle and QDrone2 aerial vehicle deliver packages from a central pickup location to customers.

COMPETITION HIGHLIGHTS

- Hands-on simulation environment in Quanser Interactive Labs (QLabs).
- Develop and test strategies using Python or MATLAB/Simulink.
- Practical delivery tasks with the QCar2 ground vehicle and QDrone2 aerial vehicle.
- Planning, coordination, and control algorithms for effective mission execution.

Registration Open Now

Register your team and begin preparing for the SMC AICA Challenge 2026.

2026 Challenge Theme: Collaborative Autonomy in Simulation

Teams will work on an autonomous package-delivery mission based on the provided scenario.

The goal is to deliver packages from a central pickup location to customers efficiently. Teams may use the QCar2, the QDrone2, or both, depending on their strategy. Successful deliveries, completion time, and window-delivery performance will contribute to each team's overall score.

Why Participate

- **$2,000 first-place prize** for the winning team.
- Visibility within the IEEE SMC community.
- Hands-on experience with next-generation autonomous systems.
- Opportunity to showcase innovative AI-driven solutions.

COMPETITION STAGES

STAGE 1

Virtual Stage

Video Submission

Teams develop and test their solutions in Quanser Interactive Labs (QLabs) using the provided Virtual Stage scenario, then submit a video link demonstrating their strategy and mission execution.

STAGE 2

On-Site Finals

October 4–7, 2026

Top teams from the Virtual Stage will be invited to compete at IEEE SMC 2026, where they will execute and demonstrate solutions based on the On-Site Finals scenario.

REGISTER TODAY

For the SMC AICA Challenge 2026, visit the official home portal for full competition information and registration details.

AICA Challenge Home Portal

Contact: smc.aica.2026@gmail.com